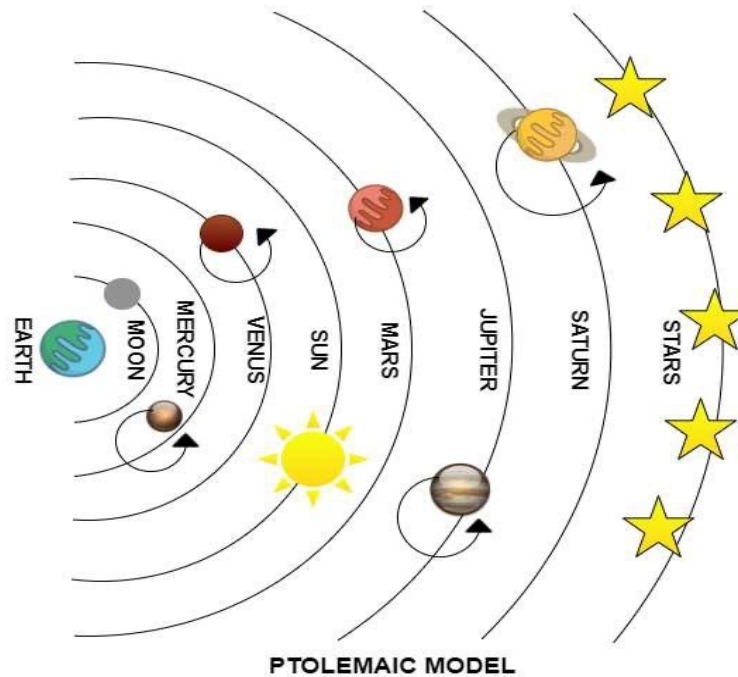


Lecture 5. Reconstructing dynamics

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Some papers/books to look at

- Chen et. al., *Neural Ordinary Differential Equations*
- Matlab and Simulink, *Dynamical System Modelling Using Neural ODE*
- Brunton, Kutz et. al, *Discovering governing equations from data by sparse identification of nonlinear dynamical systems*
- Ashwin et. al. *RNNS*

Dynamical systems

Flows:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t, \theta)$$

Maps:

$$\mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n, \theta)$$

$x(0), x_0$ Given. Study the evolution from this state

Dynamical systems link to neural networks in a number of ways

- Solving a dynamical system using a PINN .. See lecture 3
- Constructing a dynamical system operator using a Neural Operator .. See lecture 4
- Approximating the flow through a NN as a dynamical system (Neural ODE 1) [Chen et. al.] and train that dynamical system
- Using a NN to approximate/learn the RHS of a dynamical system by

$$\frac{du}{dt} = f(t, u, \theta)$$

where f is the output of a NN and learn the parameters θ from data (Neural ODE 2)

- Take f to be symbolic and learn it from data. SINDy [Brunton et. al.]
- Predict the dynamics of a dynamical system using a LSTM or similar

Neural ODES 1 [Chen et. al] : The ResNET architecture

$$x_{n+1} = x_n + \Delta t \sum_i c_i^n \sigma(A_i^n x^n + b_i^n)$$

Train this to do a specific task

Input: x_0

Output: x_N

Let $t = n \Delta t$

$$\frac{x_{n+1} - x_n}{\Delta t} = \sum_i c_i(t) \sigma(A_i(t)x^n + b_i(t))$$

Let $\Delta t \rightarrow 0$

$$\frac{dx}{dt} = \sum_i c_i(t) \sigma(A_i(t)x^n + b_i(t))$$



Neural ODE. Input: $x(0)$. Output: $x(T)$, $T = n \Delta t$



- Idea [Chen et. al.]:
- Replace the NN by solving the Neural ODE using an ODE solver eg. RK4, Adam
- Training now becomes a question of finding the functions $A(t)$, $b(t)$, $c(t)$
- Advantages in memory efficiency and back propagation
- This a problem in optimal control theory

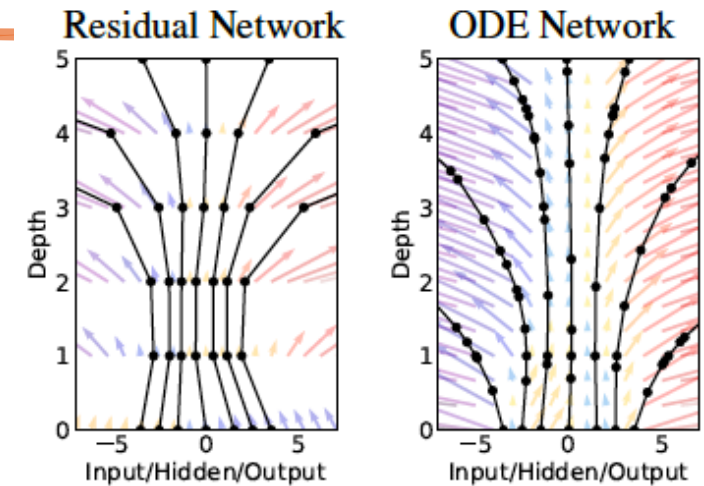


Figure 1: *Left:* A Residual network defines a discrete sequence of finite transformations. *Right:* A ODE network defines a vector field, which continuously transforms the state. *Both:* Circles represent evaluation locations.

Training such a ‘Neural ODE’. [Chen et. al 19], [Pontryagin 62]

Consider the parametrized Neural ODE

$$\frac{d\mathbf{h}}{dt} = f(\mathbf{h}(t), t, \theta), \quad \mathbf{h}(0) \rightarrow \mathbf{h}(T).$$

Compute solution using an ODE solver

Loss function $L(\cdot)$

$$L(\mathbf{z}(T)) = L \left(\mathbf{z}(0) + \int_0^T f(\mathbf{z}(t), t, \theta) dt \right)$$

To train the neural net to optimize L we need to find how it changes with θ

Results from classical control theory (and data assimilation) allow us to do this

Step 1

Define **adjoint** $\mathbf{a}(t) = \partial L / \partial \mathbf{z}(t)$ then

$$\frac{d\mathbf{a}}{dt} = -\mathbf{a}^T \partial f / \partial \mathbf{z}$$

Compute $\mathbf{a}(t)$ using an ODE solver 'backwards' starting from $\mathbf{z}(T)$



Step 2

Compute the gradient of L by solving a third ODE

$$\frac{dL}{d\theta} = - \int_T^0 \mathbf{a}(t)^T \frac{\partial f(\mathbf{z}(t), t, \theta)}{\partial \theta} dt$$

Can then train θ to optimize L using gradient descent

[Chen et. al.] implement this using an adaptive Adam solver.

Dynamical systems defined by Neural Nets and learning dynamics from data

Original DS:
$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}) \quad x \in R^n$$

Data points:
$$y_i = H(x(t_i)) + \epsilon_i, \quad i = 1 \dots N$$

Train: a NN to approximate this dynamics via:

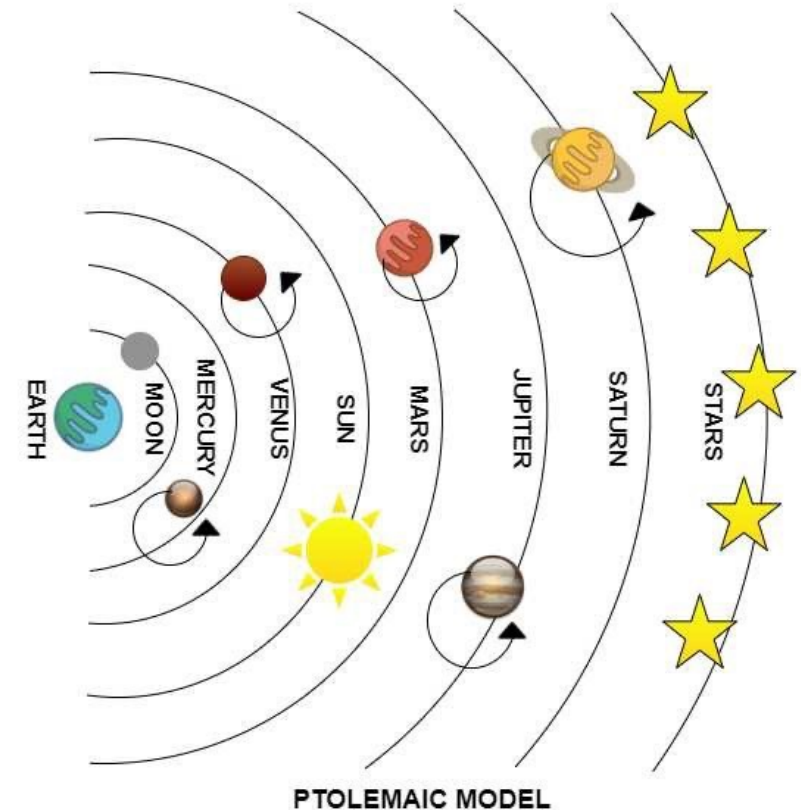
$$\frac{dz}{dt} = f(z, t, \theta) : \quad f: \text{Neural Net}$$

Also (and confusingly) called a Neural ODE eg. by Matlab



Big literature on these systems

- **Ptolemaic model** of the solar system
- Standard learning
- Echo State Networks, Reservoir networks
Especially good for chaotic dynamics
- **SINDy** approach (Kutze) using a very careful choice of RHS (can even work from video data)



Training such a ‘Neural ODE’. I [From MatLab website]

Example:

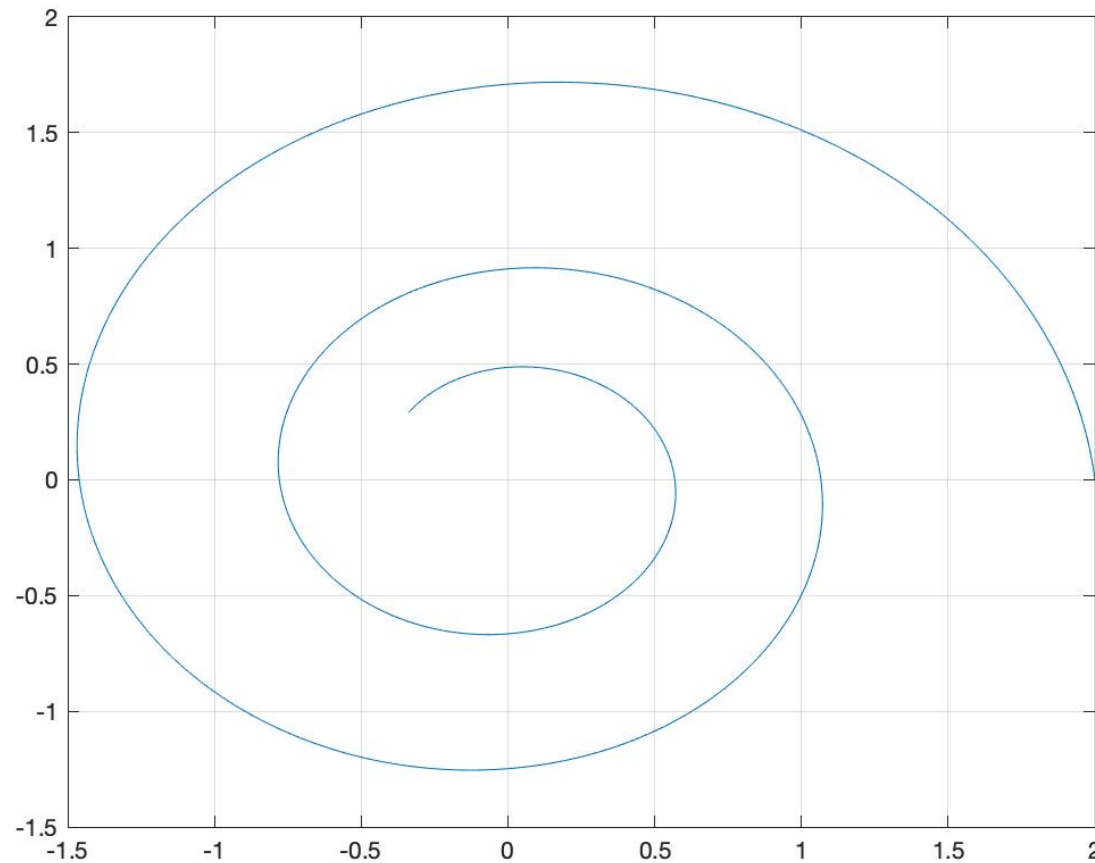
$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}, \quad \mathbf{x} \in R^2$$

$$\mathbf{x}(t) = e^{At} \mathbf{x}(0) \equiv B\mathbf{x}(0).$$

Neural ODE: $\frac{dz}{dt} = f(z, t, \theta), \quad z(0) = x(0), \quad f: \text{Neural Net}$

Solve: using RK45

Training trajectory starting from: $x(0) = [2,0]$, $t = [0,15]$, 2000 points



$$A = \begin{pmatrix} 0.1 & -1 \\ 1 & -0.1 \end{pmatrix}$$

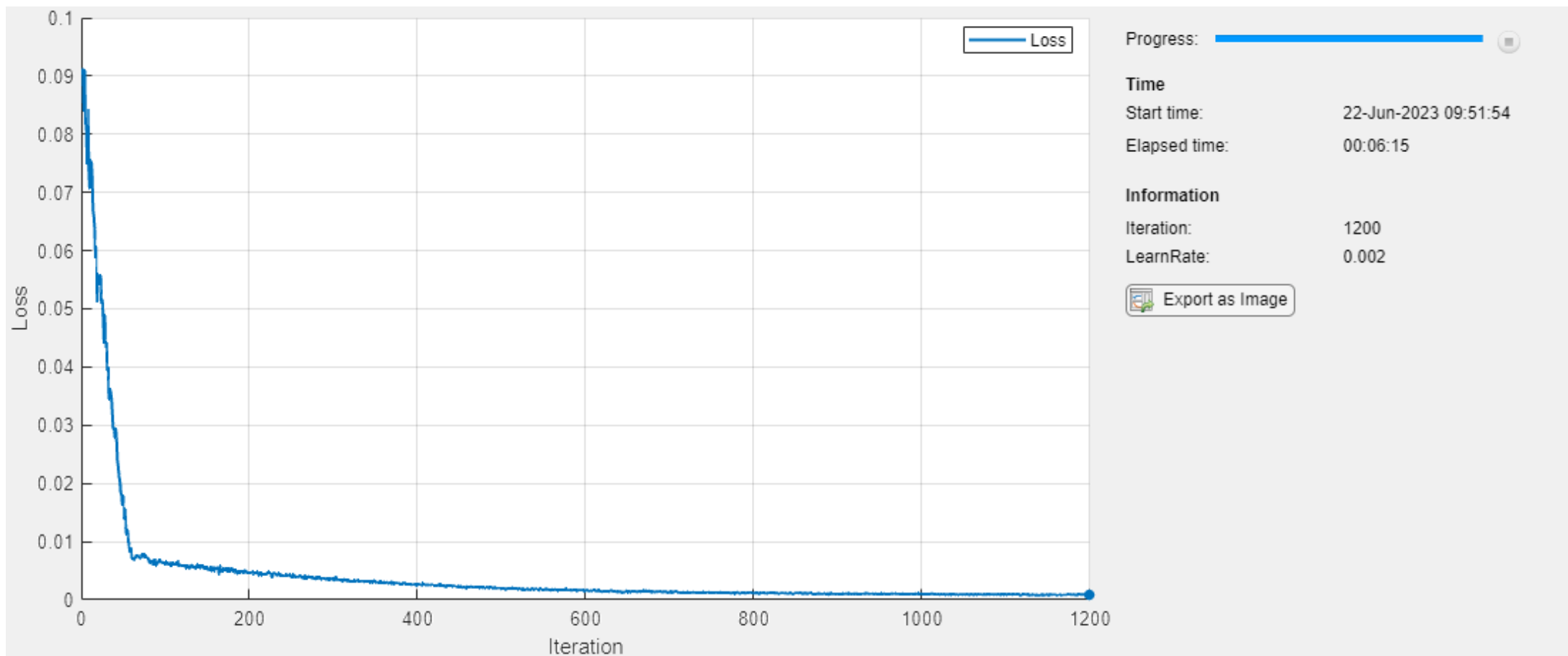
Define loss: $L(\theta)$

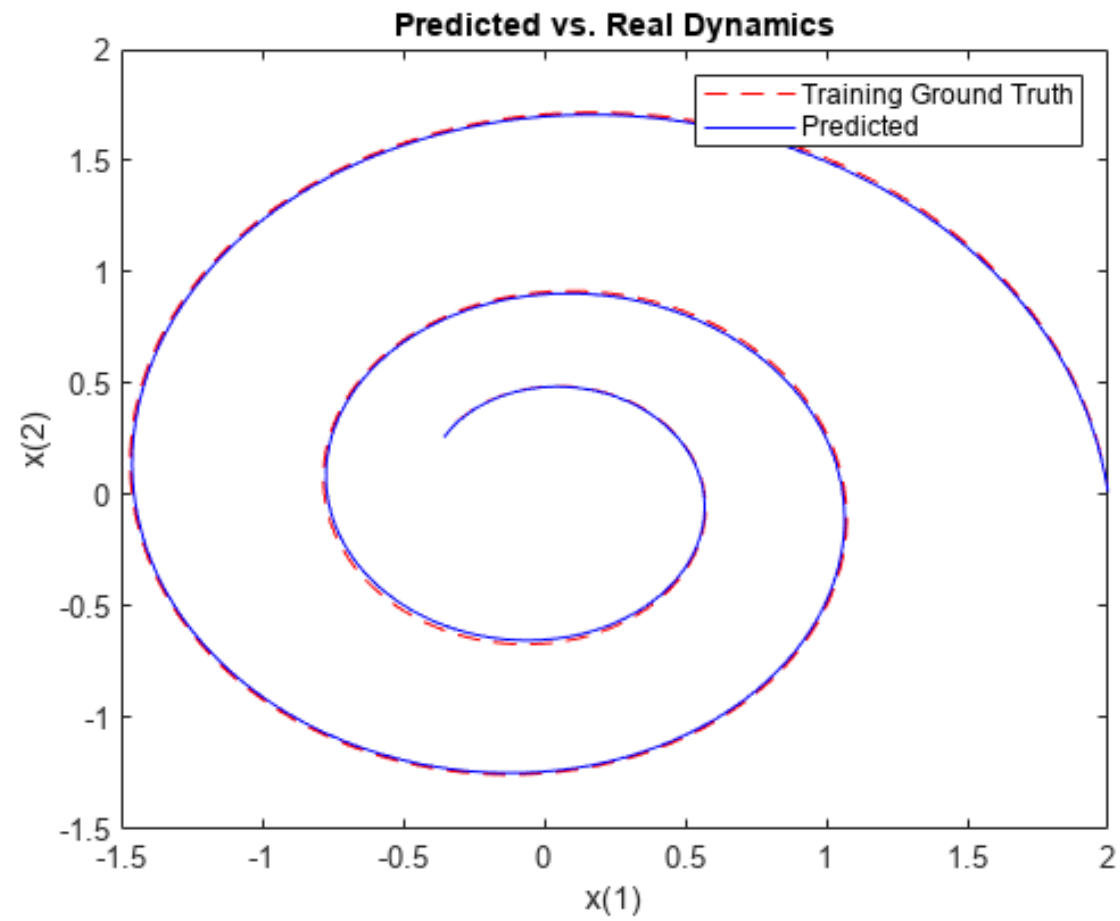
$$L(\theta) = \sum_i |H(z(t_i)) - y_i|^2$$

- Implement NN using 2 layers and tanh activation

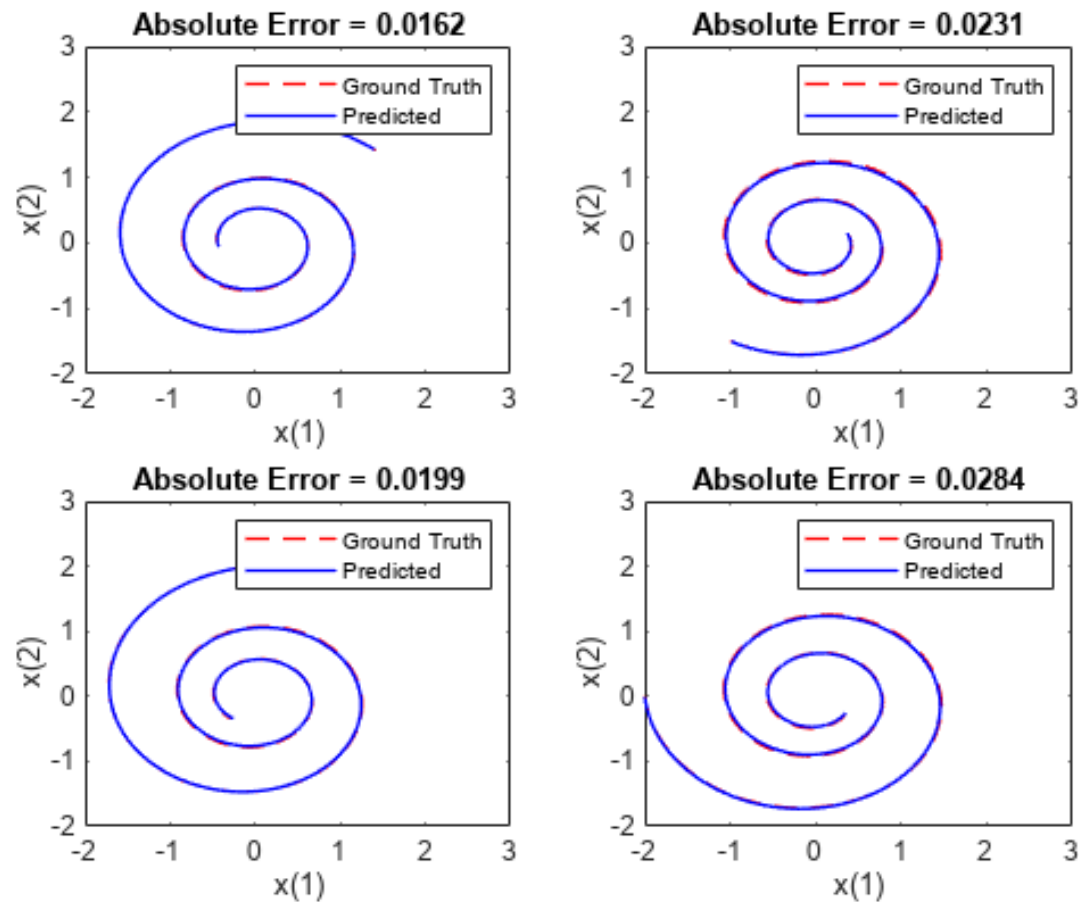
$$f(\mathbf{z}, \theta) = W_2 \tanh(W_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2.$$

- Compute gradients directly from the ODE solver using the adjoint method as before
- Optimize using ADAM using first 400 training points

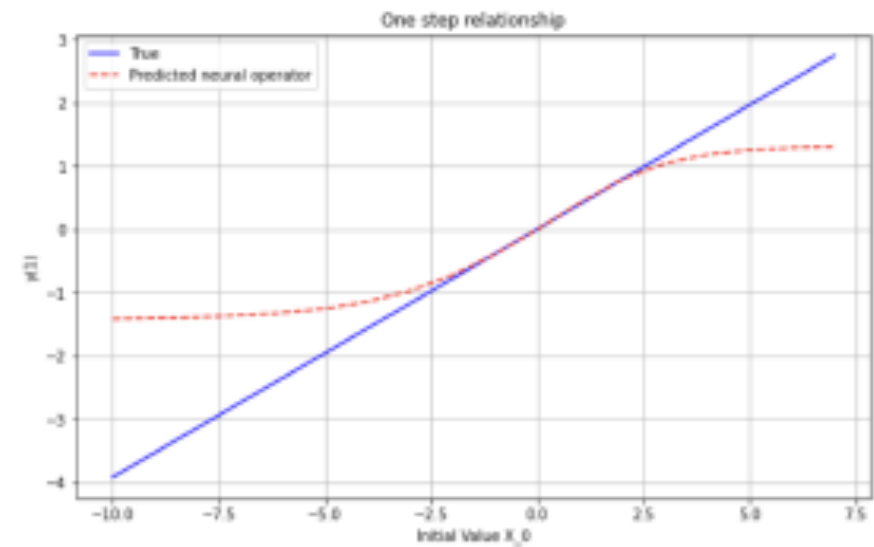
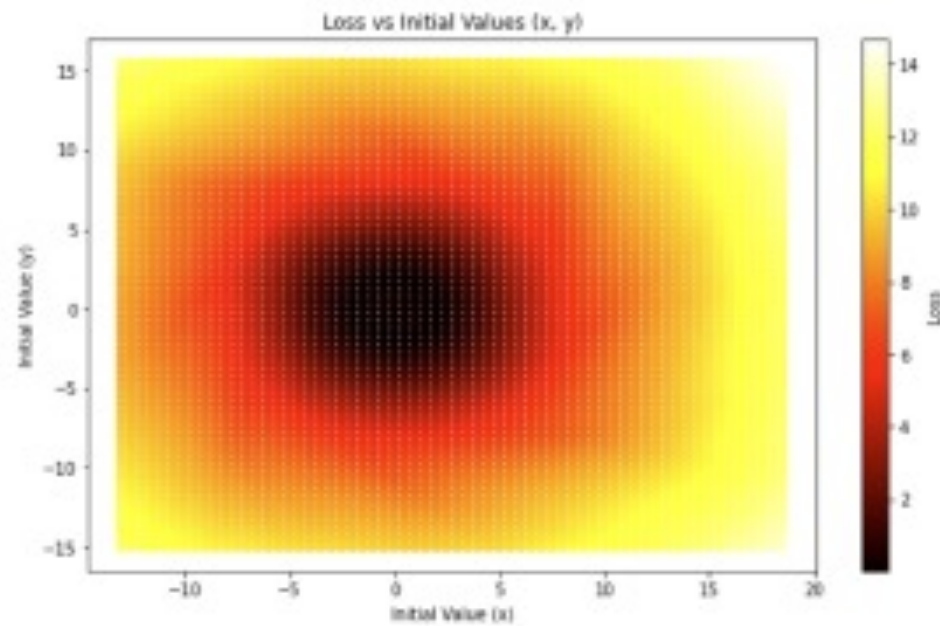




Prediction error is good for initial data close to the training trajectory



Loss increases as we move away from the training trajectory [+Libo Chen]



By way of contrast:

Assume that
$$\frac{d\mathbf{z}}{dt} = B \mathbf{z}$$

Find matrix B

100 points:

$$B = \begin{pmatrix} -0.1001 & -1.0005 \\ 1.0004 & -0.0993 \end{pmatrix}$$

400 points:

$$B = \begin{pmatrix} -0.1000 & -1.0005 \\ 1.0007 & -0.1002 \end{pmatrix}$$

2000 points:

$$B = \begin{pmatrix} -0.1000 & -1.0005 \\ 1.0005 & -0.1001 \end{pmatrix}$$



SINDy

[Brunton et. al.]

ODE: $\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x})$

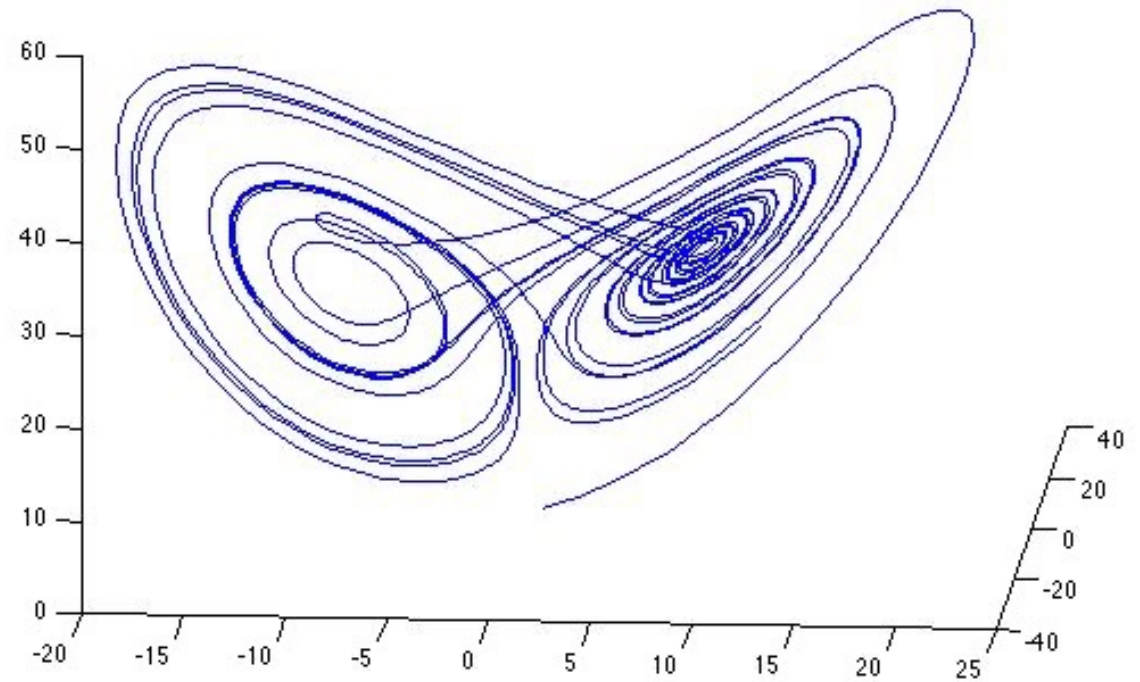
Data: $\mathbf{x}(t_i), \quad \dot{\mathbf{x}}(t_i)$ Plus noise: m measurements

Assume: $\mathbf{f}(\mathbf{x})$ can be constructed from a library of known functions

Eg. Lorenz Equations:

$r=28, b=8/3$

$$\begin{aligned} dx/dt &= \sigma(y - x), \\ dy/dt &= x(\rho - z) - y, \\ dz/dt &= xy - \beta z. \end{aligned}$$



Library: $\Theta(\mathbf{x}) = [\mathbf{1} \mid \mathbf{x} \mid \mathbf{x}^{P_2} \mid \mathbf{x}^{P_3} \mid \sin(\mathbf{x}) \mid \cos(\mathbf{x}) \mid \dots]$

Size. $m \times p$ $m \gg p$

Coefficients (sparse vectors): $\Xi = [\xi_1 \ \xi_2 \ \dots \ \xi_n]$

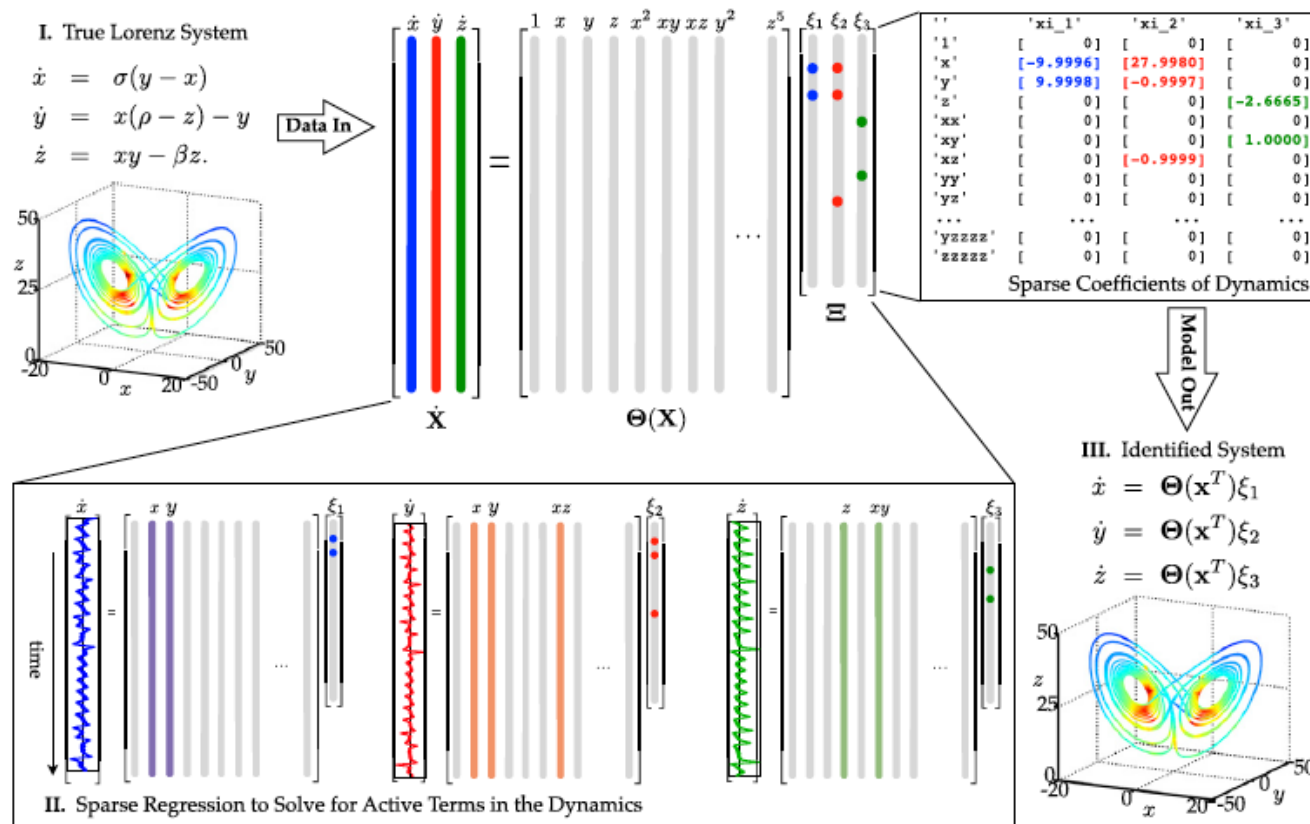
Seek a SPARSE approximation of the form:

$$\dot{\mathbf{x}} = \Theta(\mathbf{x}) \Xi$$



Aim: Find best set Ξ to fit to (filtered, noisy) data

Possibly with numerically estimated derivatives



From Brunton
et. el.

Conclusions

Neural ODEs are effective for

- Describing the flow through a NN
- Learning dynamics from data
- They inherently non smooth (or close to non smooth)
- They work best when informed by the physics